

Samsung Inovation Campus & Yeşilay & UNDP

UNHOOK: An AI-Driven Progressive Bandwidth Throttling and Adaptive Human-in-the-Loop Intervention System

Capstone Project Concept Note and Implementation Plan

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1. Project Overview

Project Summary: UNHOOK is an AI-driven digital wellness system engineered to mitigate problematic smartphone use among university students and young adults. Instead of rigid app blocking that triggers user frustration, UNHOOK establishes a personalized behavioral baseline using 4 weeks of historical usage logs collected via Machine Learning prior to installation. It then enforces progressive bandwidth throttling (local VPN packet delay) coupled with an interactive Human-in-the-Loop (HITL) intervention protocol.

Relevance to SDGs: UNHOOK directly addresses two UN SDGs:

• **SDG 3 (Good Health and Well-Being):** Mitigating physical health risks (wrist/neck musculoskeletal pain as proven by Giraldo-Jiménez et al., 2022) and mental health issues (anxiety, sleep disturbances, doomscrolling as proven by Lee et al., 2026).

• **SDG 4 (Quality Education):** Restoring cognitive focus during study sessions by reducing compulsive checking behaviors.

Problem & Impact: Compulsive smartphone habits diminish academic efficiency and physical health. UNHOOK solves this by personalizing interventions through ML-driven historical profiling and gentle, educational HITL warnings before applying friction.

2. Objectives

1. Objective 1 (Historical Baseline ML Profiling): Analyze the user's 4-week historical usage logs (app usage, unlock frequency, nocturnal ratios) using Machine Learning upon app installation to construct a highly personalized behavioral baseline profile.

2. Objective 2 (Predictive Risk Modeling): Train an machine learning risk classification engine (XGBoost/Random Forest) to compute real-time risk scores (0.00–1.00) with an F1-Score > 85%, dynamically adjusting thresholds based on individual baseline profiles.

3. Objective 3 (Human-in-the-Loop Feedback & Health Warnings): Deploy a 2-stage Human-in-the-Loop (HITL) warning system prior to restriction. The system delivers 1–2 progressive notifications explaining physical/mental health risks (e.g., wrist/neck strain, anxiety, sleep loss) and the exact rationale for throttling, adapting future interventions based on the user's immediate response.

4. Objective 4 (Adaptive Throttling & User Autonomy): Execute local VPN progressive bandwidth throttling (packet delay) for non-essential app categories when risk persists, incorporating SHAP explainability and manual override capabilities.

3. Background

Context & Necessity of Machine Learning: Existing digital wellness tools rely on static timers and hard app blocks, leading to high abandonment rates and user bypass. Crucially, generic systems fail because they evaluate screen time without knowing the user's historical habits. Machine Learning is indispensable for UNHOOK because analyzing a user's 4-week pre-installation usage history drastically improves predictive accuracy. ML enables the system to understand individual behavioral norms—such as typical study hours versus late-night bingeing—allowing UNHOOK to tailor intervention strategies to each user's unique profile.

4. Methodology

Methodological Framework: UNHOOK combines supervised Machine Learning classification with a Human-in-the-Loop (HITL) adaptive intervention protocol. During onboarding, 4 weeks of historical usage logs are processed using Machine Learning to cluster user profiles and calibrate individual baseline metrics. Real-time usage is transformed into feature vectors (AS1 nocturnal ratio, AS2 session velocity). An XGBoost model predicts real-time risk scores (0.00–1.00). When risk thresholds are crossed, the system triggers a 2-stage HITL protocol: Stage 1 delivers a health-oriented warning (explaining wrist/neck strain and sleep impact alongside the throttling rationale); Stage 2 measures the user's reaction (dismissal, app exit, or manual override) to dynamically scale VPN bandwidth throttling or feed active feedback back into the ML model.

5. Architecture Design Diagram

Component Name	Role & Core Functionality
1. Historical ML Baseline Profiler	Processes 4-week pre-installation usage logs to establish personal behavioral norms via ML clustering.
2. Real-Time Passive Log Collector	Captures Android UsageStats, screen unlocks, and notification timestamps continuously on-device.
3. XGBoost & SHAP Risk Engine	Computes continuous risk scores (0.00–1.00) and generates SHAP feature attributions based on baseline context.
4. Human-in-the-Loop (HITL) Controller	Issues 1–2 progressive health warnings (physical/mental strain explanations) and tracks user reactions.
5. Local VPN Throttling Engine	Applies progressive bandwidth throttling (packet delay) via Android VpnService to target app categories.
6. Goal, Feedback & Active Learning UI	Displays health insights, SHAP drivers, override options, and feeds user feedback back to refine the ML model.

6. Data Sources

Data Source & Preprocessing Summary (1 Paragraph Constraint): This project utilizes two complementary public dataset benchmarks: the CRAWDAD dataset (telefonica/mobilephoneuse on IEEE DataPort), containing 4 weeks of continuous, objective Android usage logs from 337 users (~297 MB) to simulate pre-installation historical baseline ML profiling, and the Pielot et al. Mobile Phone Use dataset, providing continuous smartphone engagement logs and opportunistic interaction contexts. These datasets provide granular timestamps (ts), foreground app activity (App), screen unlock events (Screen), notification interactions, and engagement metrics free from self-report bias. Preprocessing steps involve converting timestamps to local device time, filtering incomplete log days, extracting 4-week baseline profile clusters via Machine Learning, deriving real-time feature matrices (nocturnal ratios, session velocity, engagement readiness), and establishing a literature-validated proxy risk index for supervised XGBoost classifier training.

7. Literature Review

Literature Review Summary (1 Paragraph Constraint): Existing literature proves that problematic smartphone use induces musculoskeletal pain in the wrist and neck (Giraldo-Jiménez et al., 2022) as well as examination anxiety and sleep disruption in university students (Ahn et al., 2025; Lee et al., 2026), while Machine Learning models utilizing objective logs achieve high predictive accuracy (82%–97%) without requiring demographic features (Lee & Kim, 2021; Wayahdi et al., 2026). Furthermore, behavioral research demonstrates that self-nudge friction applications significantly reduce smartphone usage (Grüning et al., 2023; Olson et al., 2023), notification-disabling interventions enhance digital well-being (Dekker et al., 2025), and digital phenotyping combined with opportunistic engagement modeling provides rich context for intervention timing (Pielot et al., 2017; Schroeder et al., 2026). UNHOOK extends this literature by integrating 4-week pre-installation ML baseline profiling with a 2-stage Human-in-the-Loop (HITL) protocol—delivering 1–2 health-oriented warnings before restriction—and executing local VPN bandwidth throttling, creating an adaptive, explainable (Lee et al., 2026), and privacy-first intervention.

8. Implementation Plan

8.1. Technology Stack

• **Data Libraries & Processing:** Python (Pandas, NumPy, SciPy) for historical time-series baseline aggregation and feature matrix engineering.

• **ML Frameworks & Profiling:** Scikit-learn (LASSO, Random Forest, clustering), XGBoost, LightGBM, Imbalanced-learn (SMOTE), and SHAP.

• **Mobile & Network Components:** Flutter (Dart) for UI, Android Native (Kotlin/Java MethodChannels) for UsageStatsManager and VpnService API.

• **System & Edge AI Integration:** ONNX Runtime / TFLite for on-device inference, SQLite/Room for local storage of HITL user feedback.

8.2. Timeline & Task Distribution Matrix

Stage	Timeline	Key Tasks	Assigned Members
1. Data & Baseline ML	W1 - W3	CRAWDAD & Pielot dataset cleaning, ML baseline clustering, AS1/AS2 feature extraction, SMOTE.	Asya Ödül Koç, Kamber Can Şahin
2. Model & HITL Logic	W4 - W6	XGBoost risk model training, SHAP integration, 2-stage HITL warning logic & health messaging.	Damla Korkmaz, Damla Danış
3. Training & Feedback	W7 - W8	Cross-validation, HITL user reaction tracking, active learning loop calibration, threshold tuning.	Asya Ödül Koç, Yaren Dönmez
4. Deployment & Test	W9 - W12	Android VpnService throttling, Flutter HITL UI, local storage, end-to-end user evaluation.	All Team Members (Lead: Yaren & Asya)

8.3. Milestones

• **Milestone 1 (Baseline ML Profiling Complete):** Completion of 4-week historical baseline ML profiling and feature engineering pipeline on CRAWDAD and Pielot datasets.

• **Milestone 2 (Predictive Model Benchmark):** XGBoost risk classifier achieving Macro F1-Score > 85% with baseline-calibrated dynamic risk thresholds.

• **Milestone 3 (HITL Module Integrated):** Implementation of 2-stage HITL warning protocol delivering physical/mental health insights and tracking user reactions.

• **Milestone 4 (System MVP & Evaluation):** End-to-end Android deployment connecting baseline ML, HITL notifications, local VPN throttling, and override controls.

8.4. Challenges and Mitigations

Category	Anticipated Challenge	Mitigation Strategy
Data Quality	Variability across CRAWDAD and Pielot pre-installation usage logs.	Utilize ML clustering to group usage patterns into distinct behavioral baselines before training risk models.
Model Performance	False positive warnings causing user annoyance and premature override.	Incorporate a 2-stage HITL warning loop that learns from user overrides via active learning to dynamically adjust thresholds.
Technical Constraints	Delivering timely HITL health notifications without background OS termination.	Implement Android WorkManager and event-driven triggers on screen unlock to maintain low memory/battery overhead.

8.5. Ethical Considerations

Data Privacy & On-Device Processing: Historical baseline profiling, model inference, and HITL feedback processing occur 100% on-device (Edge AI). No raw logs or private data leave the device, ensuring GDPR/KVKK compliance.

Elimination of Algorithmic Bias: By substituting static demographic metrics with individual 4-week ML usage baselines, the system eliminates age/gender algorithmic biases.

User Agency & Health Transparency: The 2-stage HITL protocol prioritizes user autonomy by explaining restriction rationale, warning about wrist/neck/mental strain, and offering manual override options.

9. References

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